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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named “SDC2026_TeamName_Preliminary.pdf” before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	Too Good To Be True
Selected Domain	Domain 2: Orbyte Smart SME Challenge: Less Busywork, Better Business
Selected SDG Track (1–17)	SDG 8: Decent Work and Economic Growth
Project / Solution Name	eStock
Name (Leader)	Marco Ryan Suseno
Email & Phone (Leader)	hyprmarctheprime@gmail.com / +60 11-2561 2588
University Affiliation(s)	Xiamen University Malaysia
Number of Team Members	4

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Small shops must turn supplier paperwork into accurate stock records. A document may use abbreviations or package sizes that differ from the shop's catalogue. Managers understand supplier terminology, while Staff handle deliveries and daily stock movements. Retyping and passing corrections through separate messages can cause wrong product matches, quantity errors and repeated work.

A receipt listing 20 lamps does not prove that 20 usable lamps arrived. If document quantities become stock before a physical check, the business can rely on inventory that does not exist. Conversely, requiring the Manager to repeat every Staff action adds administrative work.

eStock focuses on micro and small shops moving from paper or spreadsheets to one inventory record for one warehouse. Our challenge is to reduce repetitive document entry while preserving the distinction between expected and physically received goods.

A May 2026 Khazanah Research Institute paper describes fragmented digital adoption and back-end processes that remain manual among Malaysian micro and small enterprises. This supports the broader productivity problem; our specific receiving workflow still needs validation with business users.

Mentor feedback helped us narrow an earlier forecasting idea toward practical stock entry. We have not established customer adoption, measured business losses or real-store time savings. These will be investigated through controlled synthetic tasks and subsequent user research.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

eStock is a document-assisted inventory application for one shop and warehouse. A Manager photographs or uploads a supplier receipt, invoice or delivery note. Google Gemini 3.8 Flash extracts document type, date, product descriptions, quantities, units and review cues into a structured draft. Application logic suggests catalogue matches using official names, confirmed aliases and conservative similarity rules.

The Manager reviews the draft, resolves unfamiliar products and manually converts expected quantities into base units. Sending freezes the reviewed receipt and notifies Staff without increasing stock.

Staff checks physical goods and enters actual received quantities. The backend calculates shortages or extras and posts only the actual quantities. Inventory, movements, receipt completion and required discrepancy notifications are saved together, with protection against repeated posting. The Manager receives an informational alert; no further approval is required. For example, a document listing 20 lamps with only 18 received adds 18, not 20.

Both roles search inventory and record Take or usable Return. Server rules prevent withdrawals exceeding available stock. Manager controls cover product management, drafts and History; destructive clearing actions require confirmation.

The implementation uses Next.js, React, TypeScript, PostgreSQL and the Google Gen AI SDK, with Replit deployment and Docker-based local development. It applies the GenAI API, relational database and cloud-hosting skills covered by the Phase 1 workshops.

AI interprets variable document content; application rules and human physical checks control stock changes. Images are transient application inputs; saved records contain structured fields. Competition demonstrations and benchmarks use team-created synthetic data. Exploratory live OCR checks inform improvements, while general accuracy, processing speed and hosted durability remain evaluation tasks.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Component: Technology and purpose

Web interface: Next.js 16.4.2, React 19.3.1 and TypeScript 6.0.3 provide Manager and shared Staff interfaces. Plain CSS and Manrope support a clear, responsive presentation.

Backend: Next.js server routes on Node.js enforce sessions, permissions, input validation and inventory operations. Credentials remain server-side.

AI document extraction: Google Gemini 3.8 Flash, configured as gemini-3.8-flash, is called through the Google Gen AI SDK. It extracts document information into structured, reviewable fields.

Product matching: Application logic normalizes names, checks official catalogue names and confirmed aliases, and conservatively ranks similar descriptions. Ambiguous matches require Manager selection. Aliases do not retrain Gemini.

Database: PostgreSQL, accessed through pg, stores products, aliases, drafts, frozen submissions, physical receiving results, stock movements, notifications and session records.

Stock integrity: Database transactions, product locking, receipt revisions and persistent request identities protect against partial updates, stale checks, negative stock and repeated posting.

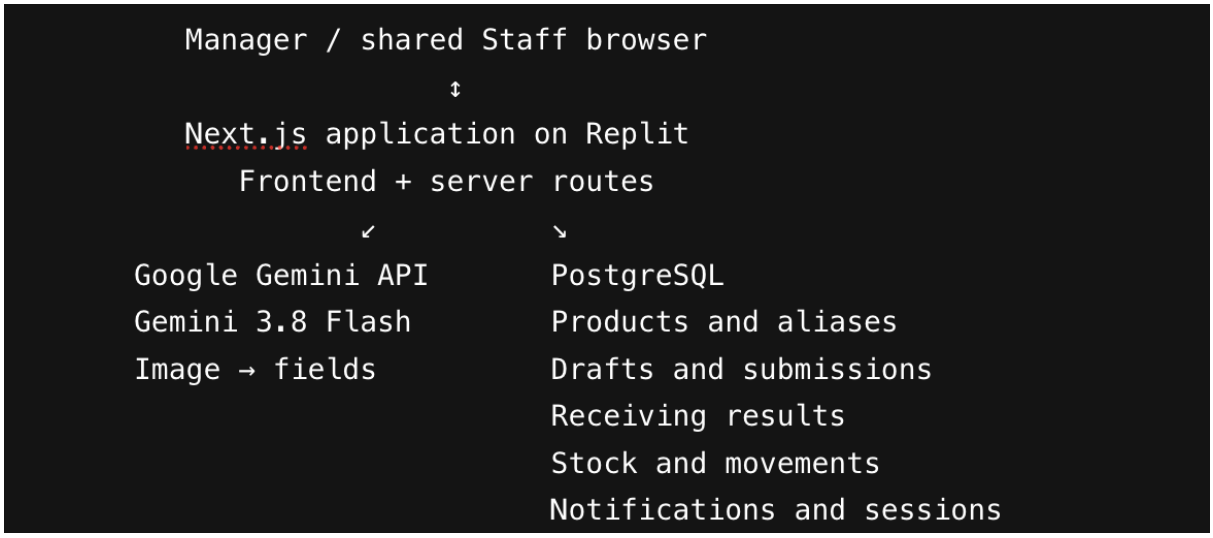
Local development: Docker Compose supplies an independent PostgreSQL database with a persistent volume on each team member's laptop.

Deployment: Replit is used for application deployment. The final published revision, public access and hosted PostgreSQL persistence must be checked against the submitted build.

Verification: TypeScript, Biome, production builds, automated PostgreSQL/API tests and Playwright-driven Chrome/Edge checks support implementation verification.

Orbyte alignment: The workflow follows role permissions, connected business information, human decisions, notifications and visible operational records. A working Orbyte API integration has not been established.

System architecture and component connections



The browser communicates with the application backend. Only the backend connects to Gemini and PostgreSQL.

During scanning, the backend sends a document image to Gemini, validates the returned fields and uses catalogue information to prepare the draft. Manager corrections and sending create a frozen submission without adding stock.

Staff confirmation sends actual received quantities to the backend. One database transaction records the receipt results, updates stock, writes movements and creates required discrepancy notifications. The interface then displays the saved results. Gemini has no direct database access or stock-write authority.

Official sponsor resources used

Yes.

Replit: Used for eStock deployment. The team's workspace is MaximillionC/eStock. Final published-build access and restart/redeployment persistence require verification.

RunPod: Used earlier in the hackathon for Chronos-2 forecasting training and evaluation on an NVIDIA GeForce RTX 4090. This historical experiment is declared separately and does not power the current receipt-processing workflow.

Alpha Red Solutions / Orbyte: Domain materials and mentorship informed problem refinement and workflow design. The team made the final product decisions and implemented the solution.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Mentor feedback challenged our original forecasting idea: another inventory tool could create duplicate entry. We narrowed the scope to one shop establishing one warehouse's records and designed around the Manager's product knowledge and Staff's physical checks.

The prototype extracts document details, supports Manager corrections and product matching, and freezes expected quantities before sending to Staff. Staff enters actual counts; confirmation posts stock once and alerts the Manager to discrepancies.

Local verification covers database/API safety and Chrome/Edge journeys, including permissions, retries, receiving, responsive layouts and failure states. Earlier exploratory Gemini checks used selected real, team-provided supplier-document photos, some containing personal data; they exposed slow responses and plausible extraction errors. Competition demonstrations and benchmarks use team-created synthetic documents.

Replit has been used for deployment. We still need to verify the final public build, judge access and database persistence across restart/redeployment. Next, identical synthetic receipts will compare manual and assisted completion times, correction counts and final stock correctness. The numeric baseline and percentage target are planning assumptions, not achieved results.

Before a business pilot, we will validate usability with owners and Staff and review provider retention, access controls and backups.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	SDG Target 8.2: Improve productivity through technological upgrading and innovation. eStock targets repetitive document entry and stock-recording work. SDG Target 8.3: Support entrepreneurship and the development of micro, small and medium-sized enterprises. eStock aims to make structured inventory recording more practical for small shops. These are project-level contributions, rather than claims of measured national economic impact. Source: United Nations, Goal 8.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	Estimated planning baseline, not a measured result: 10 minutes to manually enter and reconcile one synthetic supplier document containing 10 product lines, including product identification, quantity recording and Staff verification. This is a team planning assumption, not a published industry statistic. Before reporting achieved improvement, we will replace it with the measured median from comparable manual tasks.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	Median end-to-end processing time, in minutes, per synthetic 10-line document, from the start of document entry to completed Staff confirmation and correct stock posting. Percentage improvement = ((manual median time − assisted median time) ÷ manual median time) × 100. Supporting measures are correction counts, incorrect final balances and duplicate stock postings. Both workflows will

Metric Category	Your Response
	use equivalent catalogue information and delivery scenarios; assisted timing includes AI waiting time.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	Target, not an achieved result: At least a 30% reduction in median completion time compared with the measured manual baseline, while maintaining correct inventory and preventing duplicate posting in the benchmark cases. Against the provisional 10-minute baseline, this would correspond to 7 minutes or less. Evaluation will use at least 10 team-created synthetic documents covering matching deliveries, shortages, extras and product-name variations. Actual results will be reported even if the target is not reached.

Our primary intended contribution is operational productivity under SDG 8. Environmental benefits, such as reduced paper use or avoided waste, will not be quantified without supporting measurements.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- [V] All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- [V] The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- [V] The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- [V] All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
Next.js	Code / web framework	MIT
React and React DOM	Code / UI libraries	MIT
Node.js	Code / runtime	MIT and bundled third-party notices
TypeScript	Code / development tooling	Apache License 2.0
node-postgres (pg)	Code / database client	MIT
PostgreSQL	Code / database	PostgreSQL License

Google Gen AI SDK (@google/genai)	Code / API SDK	Apache License 2.0
Google Gemini 3.8 Flash	Pre-trained model accessed through an API	Google APIs and Gemini API terms
Manrope through Fontsource	Font asset	SIL Open Font License 1.1
Biome	Code / development tooling	MIT OR Apache License 2.0
Playwright Core	Code / browser verification	Apache License 2.0 and applicable NOTICE
Development type-definition packages	Code / development tooling	MIT
Docker Compose	Code / local development tooling	Apache License 2.0
Docker Desktop	Development software	Docker Subscription Service Agreement
Python	Code / development and historical training runtime	Python Software Foundation License and distribution notices
SQLite through Node's SQLite support	Code / preserved earlier implementation	SQLite public-domain dedication and Node distribution notices
Chronos-2, historical forecasting experiment	Pre-trained model	Apache License 2.0
chronos-forecasting, historical forecasting experiment	Code / model library	Apache License 2.0

AutoGluon, historical forecasting experiment	Code / machine-learning framework	Apache License 2.0
PyTorch, historical forecasting experiment	Code / machine-learning framework	BSD-style licence and accompanying notices
RunPod Python SDK, historical work	Code / cloud SDK	MIT

Datasets and reference material

Current competition demonstrations and benchmark cases use team-created synthetic inventory records and document images.

Selected real, team-provided supplier-document photos containing personal data were sent to Gemini for exploratory OCR QA. They are excluded from competition demonstrations and benchmarks and are not reproduced here. Permission and retention arrangements remain unresolved.

AI-assisted development

The team used OpenAI Codex with GPT-6.1 Sol, ChatGPT, and Anthropic Claude with Claude Opus 5.5 for development and project-material assistance. These tools are declared separately from the Gemini model used by the application. Applicable provider terms govern their use; the team remains responsible for reviewing and explaining the submitted work.

The project's Apache License 2.0 does not replace third-party licenses, notices or service terms.

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

Name: Marco Ryan Suseno

Signature: 

Date: 25 – 10 - 2026